Multi-channel Source Separation 1-1

안인규 (Inkyu An)

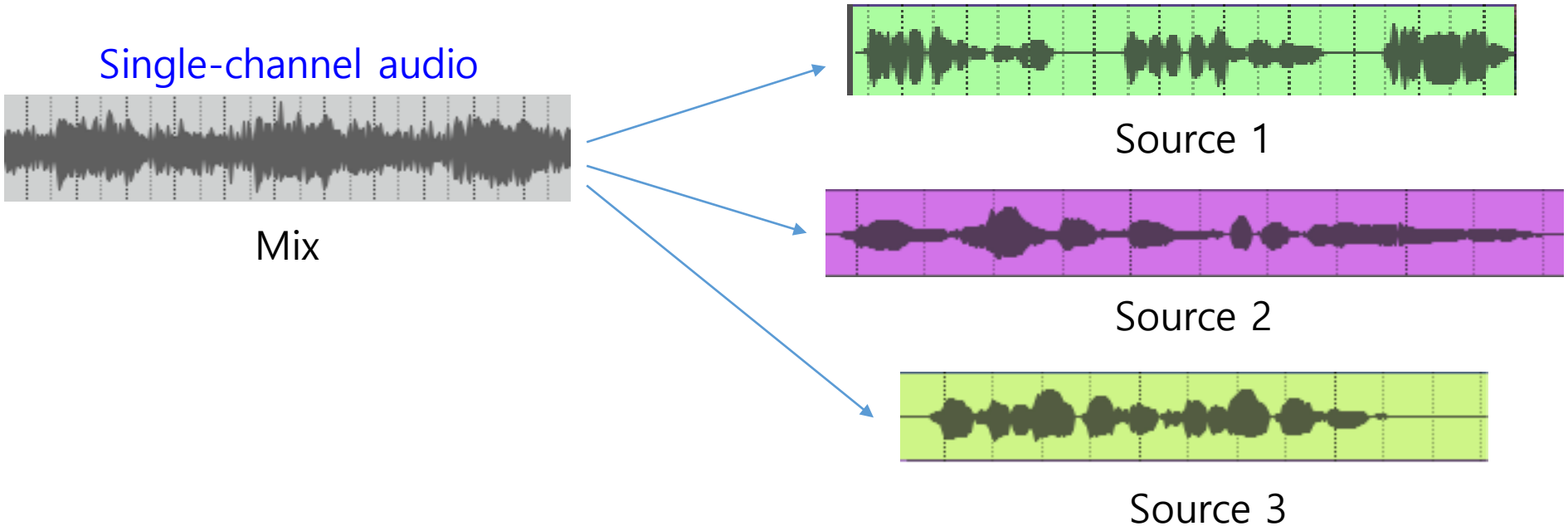
Speech And Audio Recognition
(오디오 음성인식)

https://mairlab-km.github.io/

This lecture material refers to
https://github.com/yandexdataschool/speech_course?tab=readme-ov-file and
https://github.com/markovka17/dla

Blind Source Separation

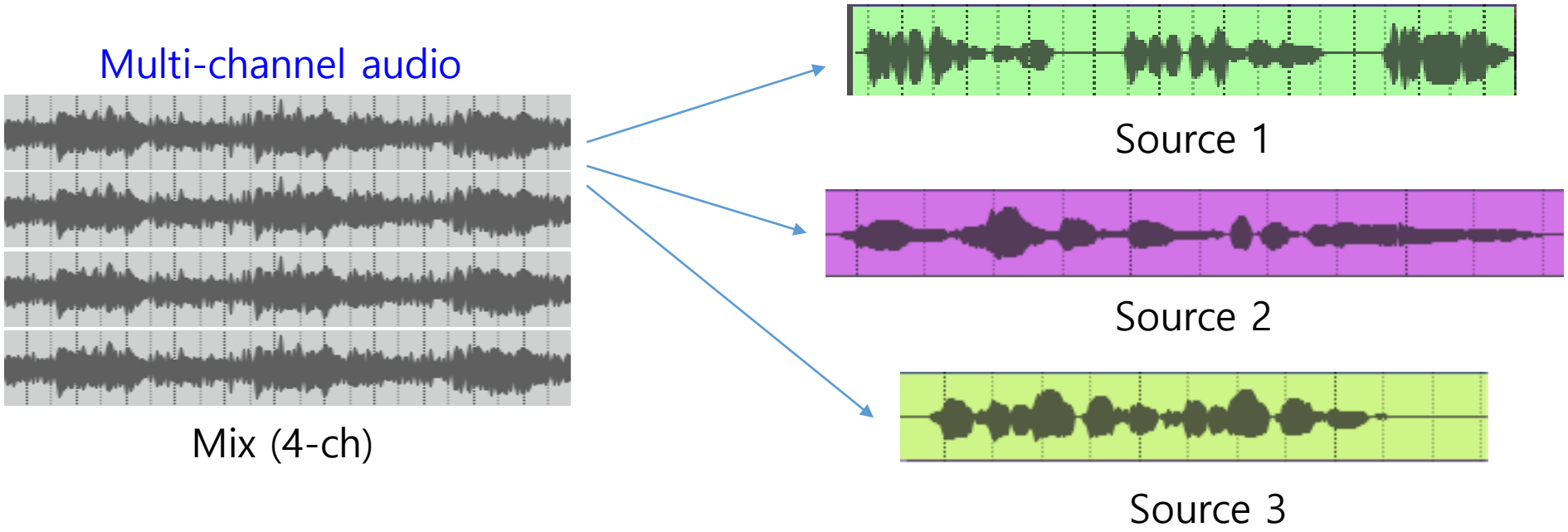
- **Goal**: extract K sources from the noisy mixture **w/o (or with very little) information about the mixing process**



- We know denoising and specific guided separation, how can we apply DL here?

Multi-channel Source Separation

- **Goal**: extract K sources from the noisy mixture of multi-channel audio

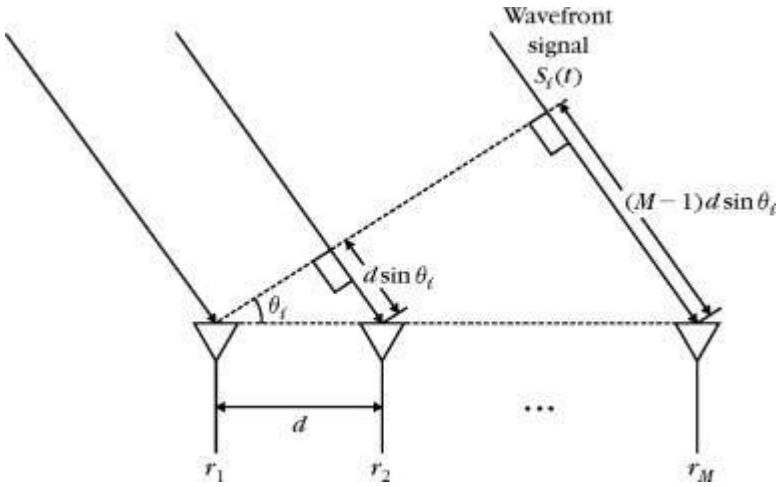


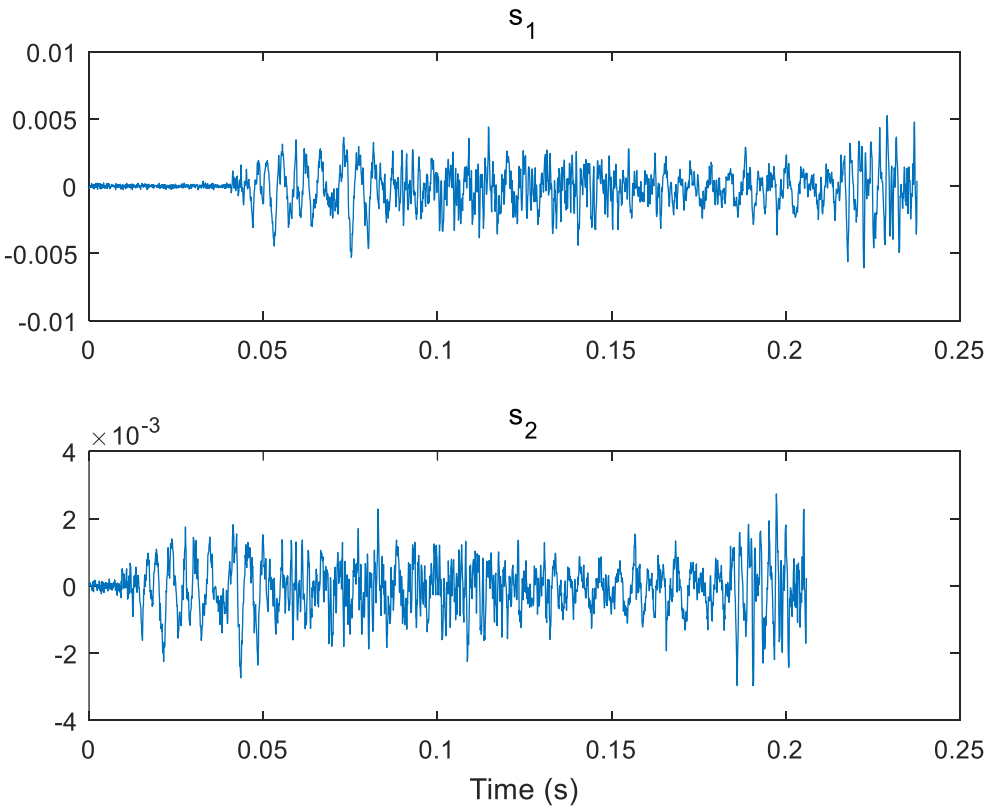
- How can we measure the multi-channel audio signal?

※ Today: DoA-based spatial filtering (not fully blind) — blind ICA/IVA at the end

Multi-channel Source Separation

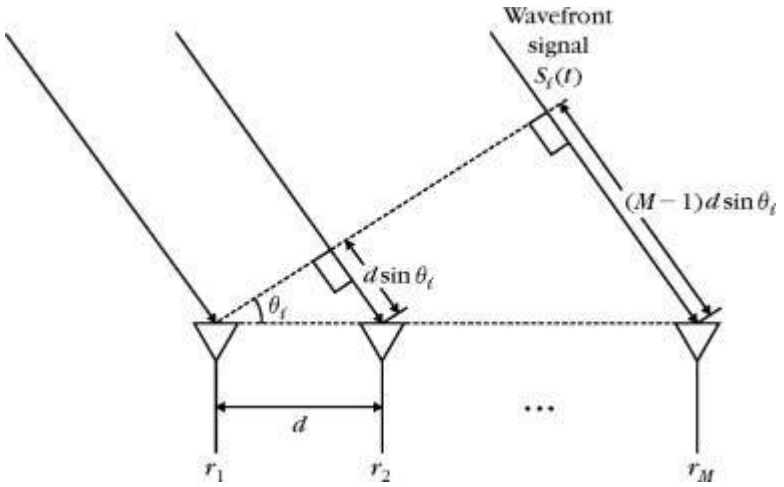
- Microphone Array
  - Consider the Uniform Linear Array
  - What's different about the audio collected at each microphone?

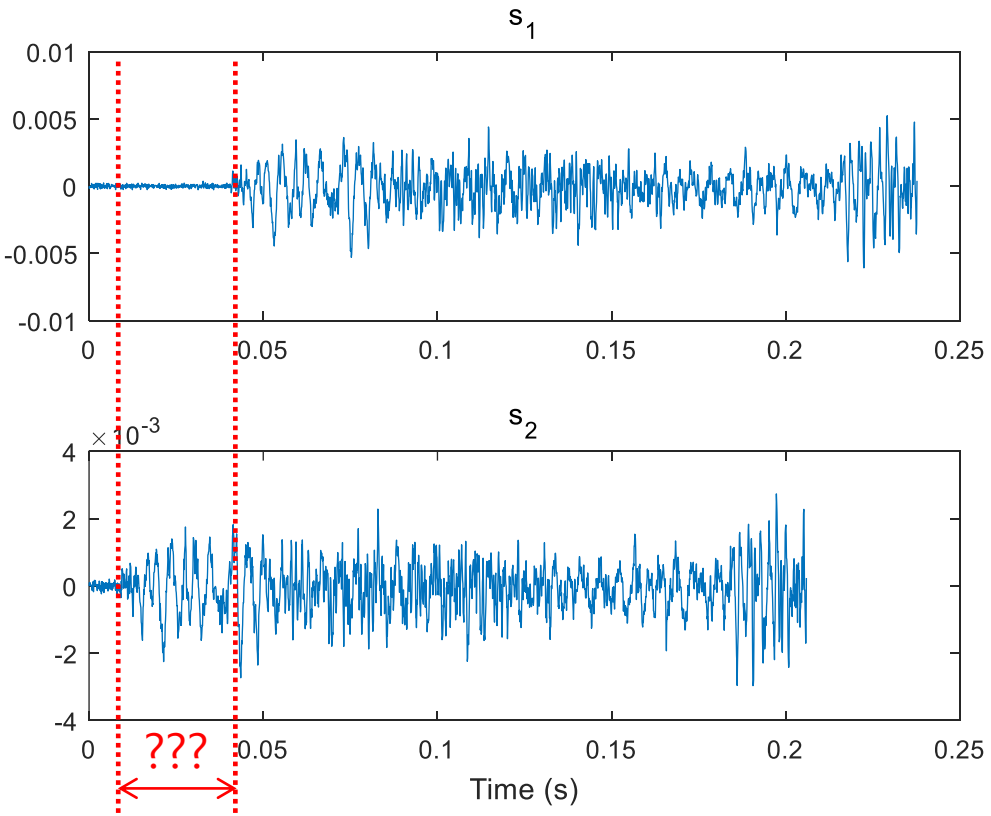




Multi-channel Source Separation

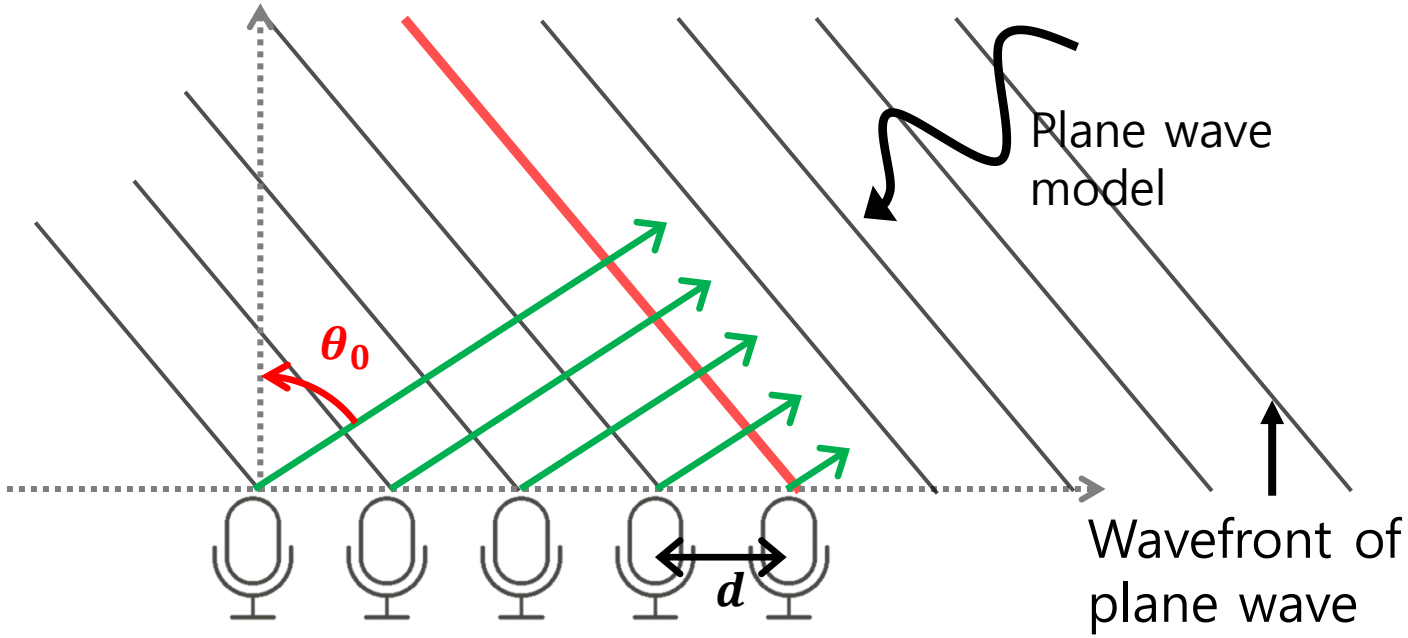
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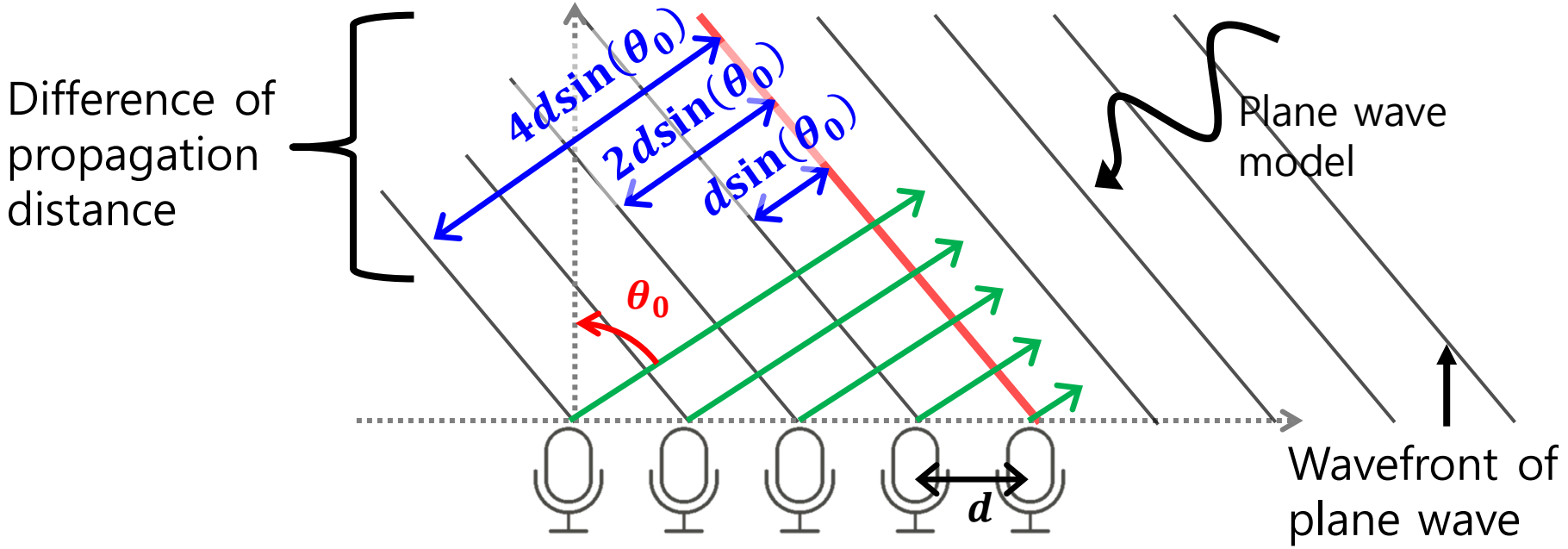
Multi-channel Source Separation

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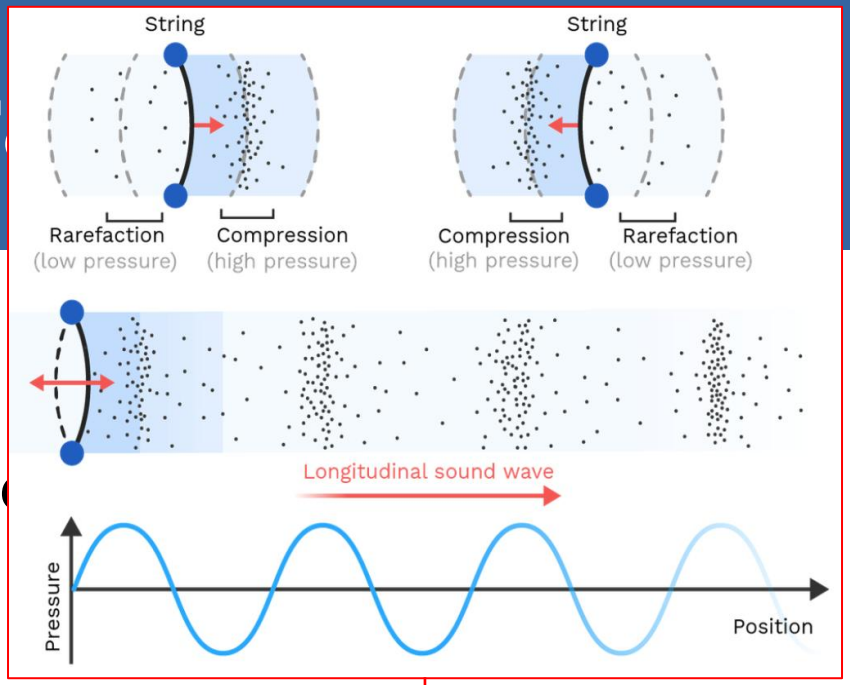
Multi-channel Source Separation

- Microphone Array
  - Consider the Uniform Linear Array
  - What's different about the audio collected at each microphone?

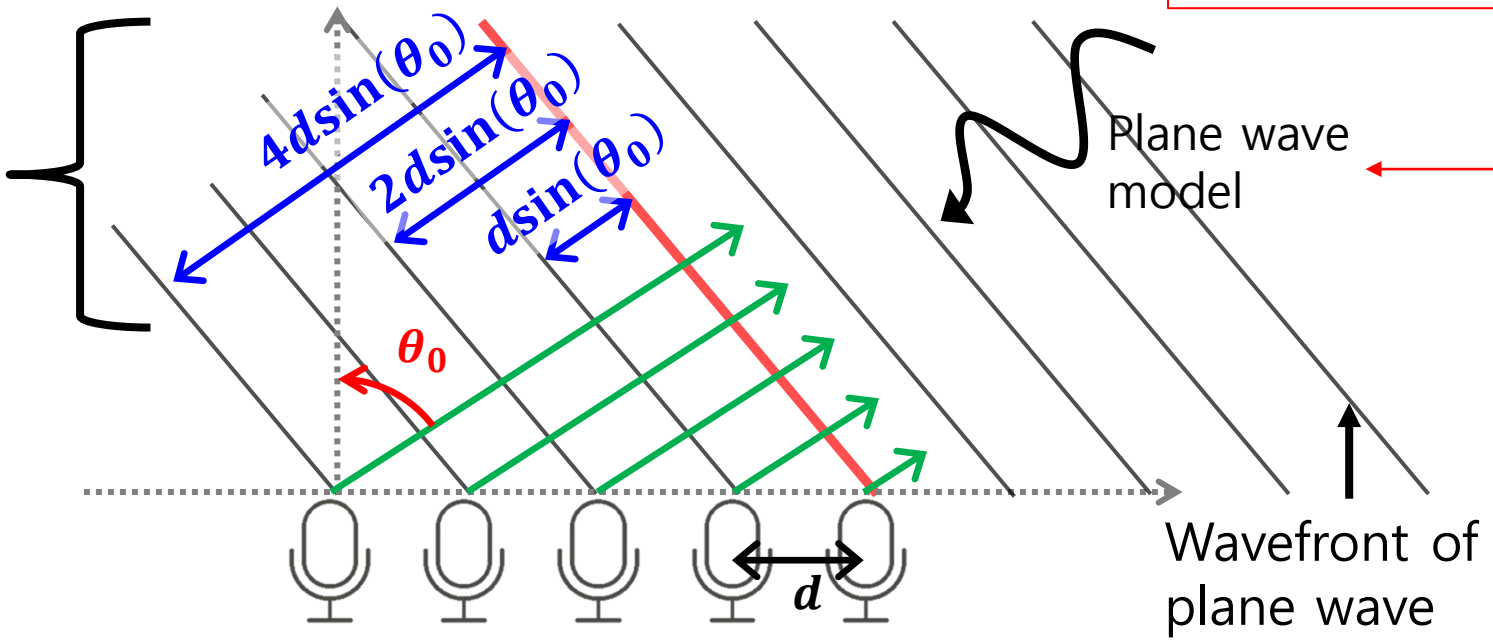


Multi-channel Source Separ

- Microphone Array
  - Consider the Uniform Linear Array
  - What's different about the audio collected at ea



Difference of propagation distance



Time delay at mic m:

$$\tau_m = \frac{(m-1)\, d\sin\theta_0}{c}$$

c ≈ 343 m/s

Multi-channel Source Separation

- Microphone Array
  - Is it possible to extract only the audio signal coming from a specific direction? → **Beamforming**
  - What is the beamforming algorithm?
    - Focusing a specific direction, $\boldsymbol{\theta}$, by using beamforming weights (spatial filter)

$$y(\theta,t)=\sum_{m=1}^{M} g_m(\theta,t) * x_m(t) \quad \Longleftrightarrow \quad y(\theta,\omega)=\boldsymbol{w}(\theta,\omega)^H \boldsymbol{x}(\omega)$$

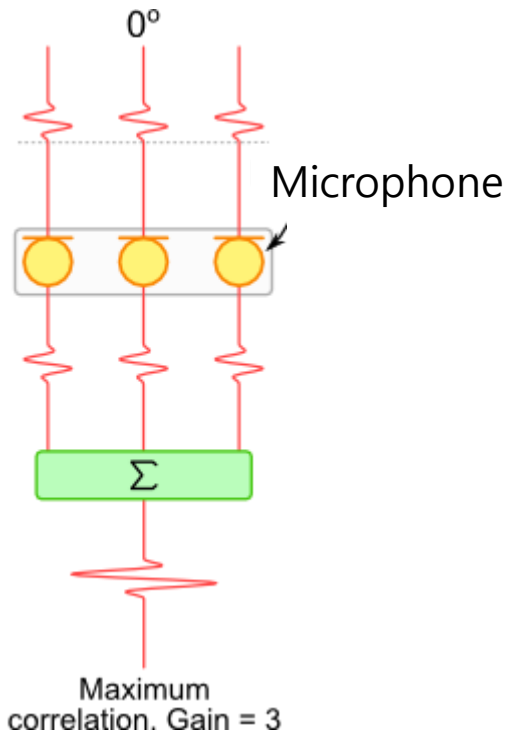
Convolution

FFT & iFFT

Beamforming output | Weight (Spatial filter) | Measured pressure signals

Multi-channel Source Separation

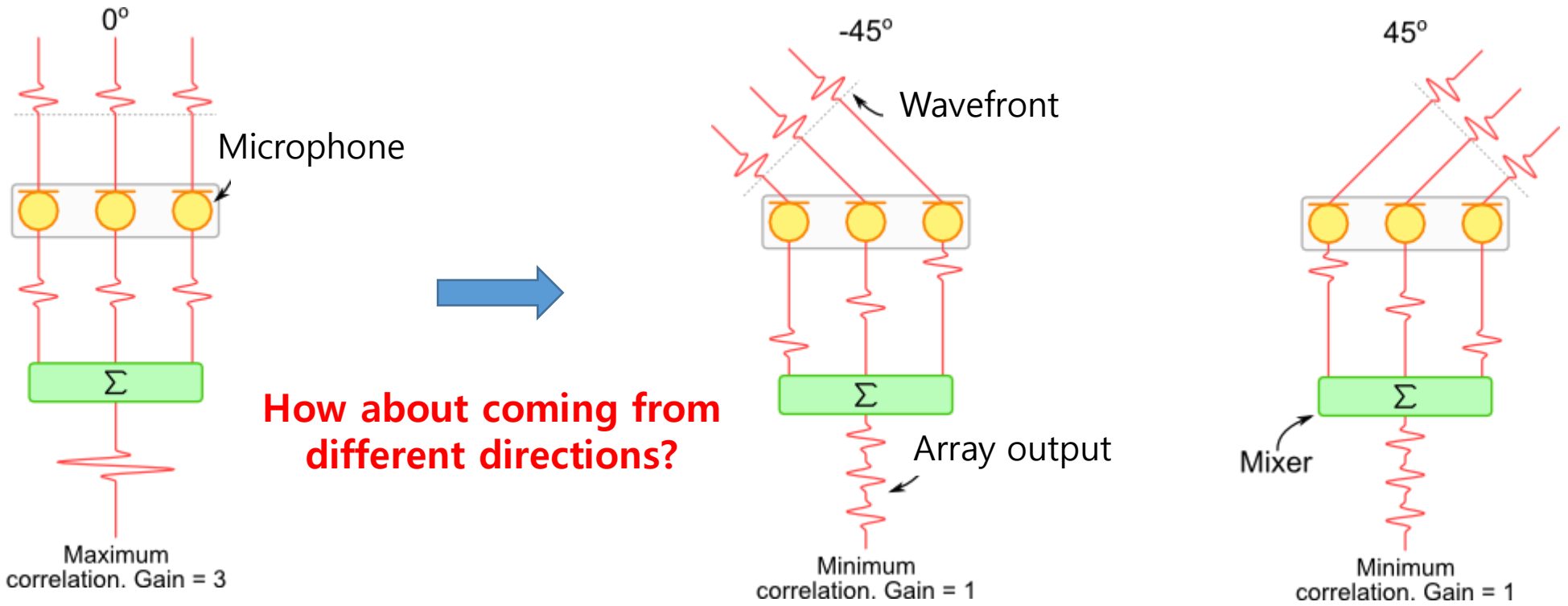
- Microphone Array
  - Is it possible to extract only the audio signal coming from a specific direction? → **Beamforming**
  - A simple method? → **Add all the multi-channel signals together**



https://medium.com/@itberrios6/introduction-to-beamforming-part-1-92767408f483

Multi-channel Source Separation

- Microphone Array
  - Is it possible to extract only the audio signal coming from a specific direction? → **Beamforming**
  - A simple method? → **Add all the multi-channel signals together**



https://medium.com/@itberrios6/introduction-to-beamforming-part-1-92767408f483

Multi-channel Source Separation

- **What we want**: not just a weighted sum, but a filter

Weighted sum (previous slide)

$$y[n] = a \cdot x[n]$$

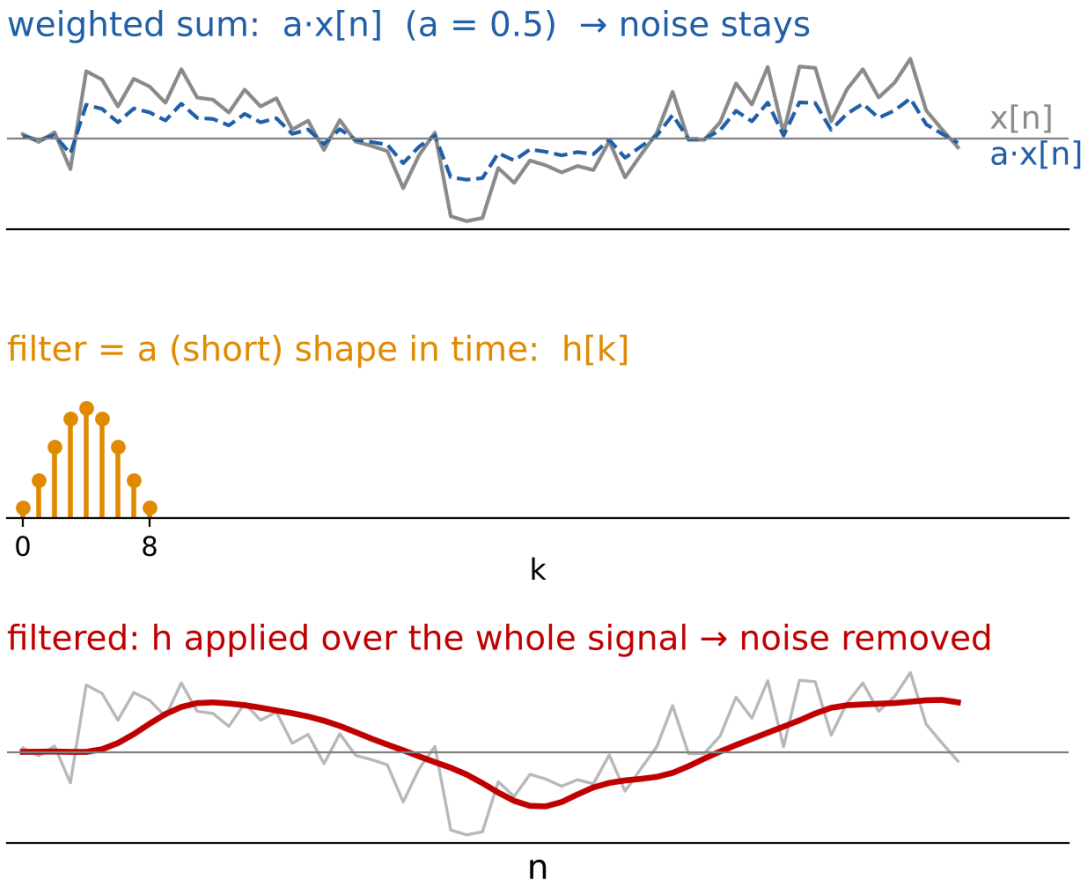
one number per channel → everything is scaled equally

Filter

$$h[k], \quad k = 0,1, \ldots, K-1$$

a shape in time, applied over the whole signal

The shape of h decides what is kept (signal), removed (noise), or delayed



Multi-channel Source Separation

- Applying the filter to the whole signal = convolution

At each time n

$$y[n] = h[0]\,x[n] + h[1]\,x[n-1] + \cdots$$

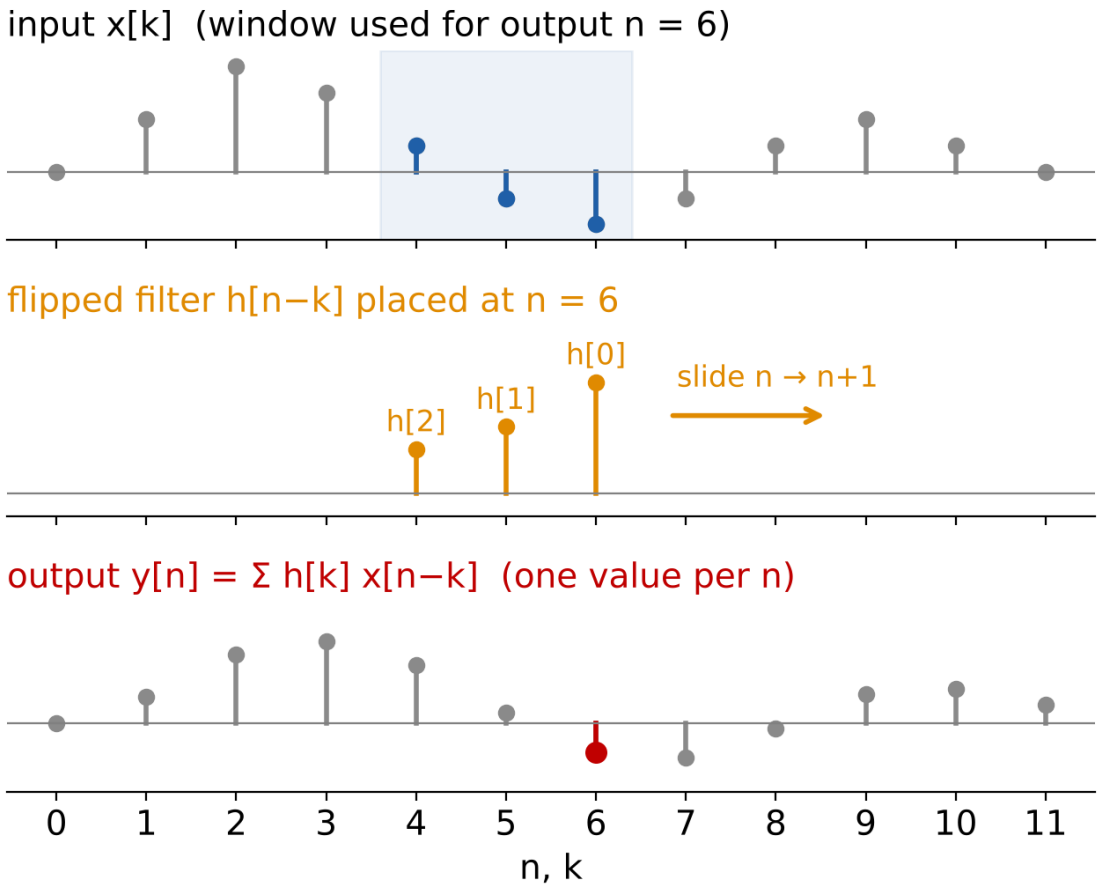
weighted sum of the current and past inputs, weights = h

For every n → convolution

$$y[n] = \sum_k x[k]h[n-k] = (h * x)[n]$$

Continuous time

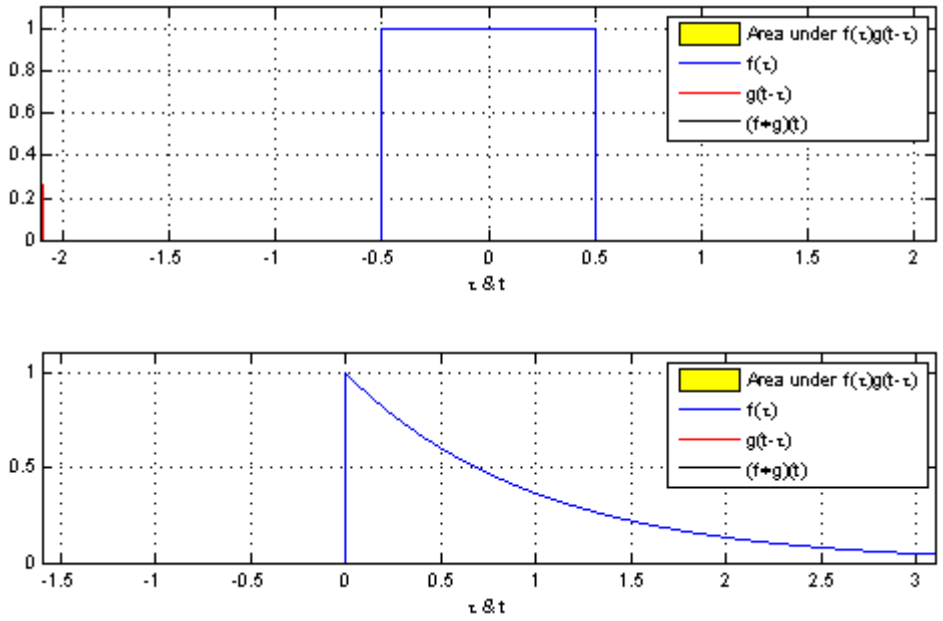
$$y(t) = \int x(\tau)\,h(t-\tau)\,d\tau$$

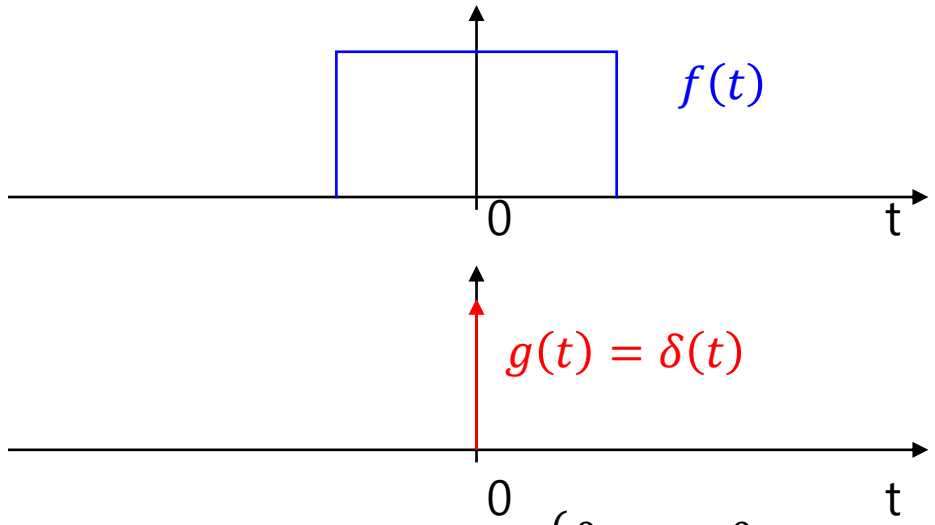


Multi-channel Source Separation

- Microphone Array
  - Convolution: $(f * g)(t) := \int_{-\infty}^{\infty} f(\tau) g(t - \tau)\, d\tau.$

Example of Convolution





Dirac delta function $\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$

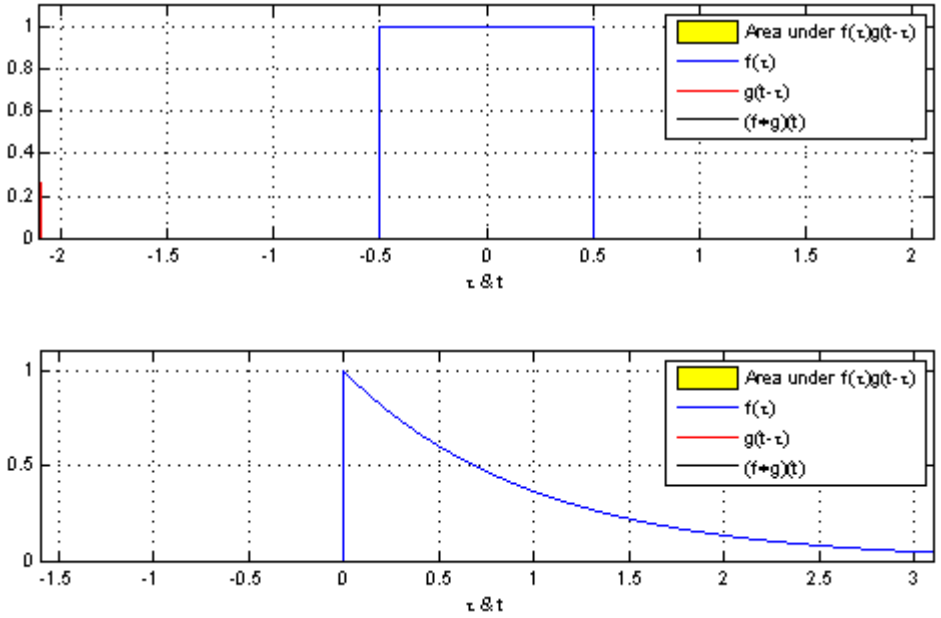
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

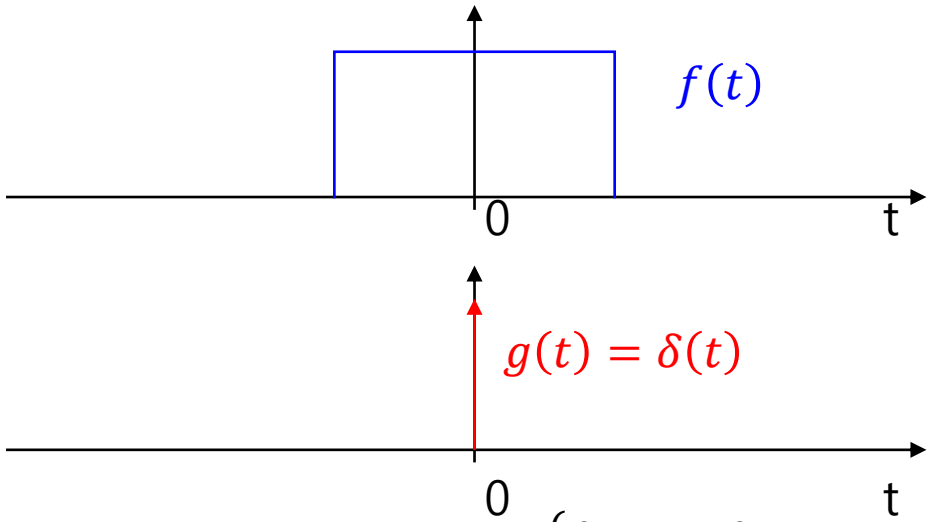
$(f * g)(t)$?

Multi-channel Source Separation

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Example of Convolution

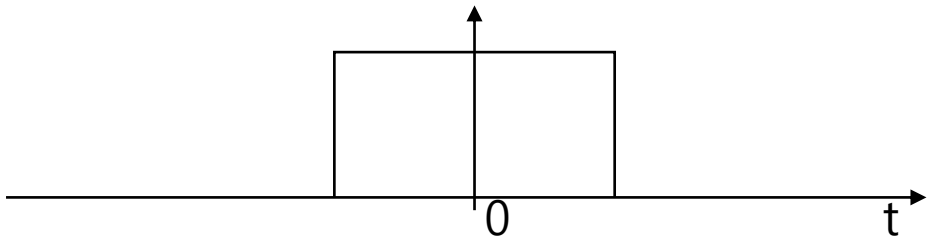




Dirac delta function $\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

$(f * g)(t)$?



Multi-channel Source Separation

- LTI system = Linear + Time-Invariant



Linearity

$$a\,x_1[n] + b\,x_2[n] \;\rightarrow\; a\,y_1[n] + b\,y_2[n]$$

scale & add inputs → outputs scale & add

Time-invariance

$$x[n - n_0] \;\rightarrow\; y[n - n_0]$$

delay the input → output is just delayed

Impulse response

$$\delta[n] \;\rightarrow\; (h * \delta)[n] = h[n]$$

unit impulse in → the filter shape h comes out (continuous time: Dirac delta δ(t))

Room acoustics (over a short time) and a fixed-weight beamformer can be modeled as LTI

Multi-channel Source Separation

- Why does an LTI system compute a convolution?

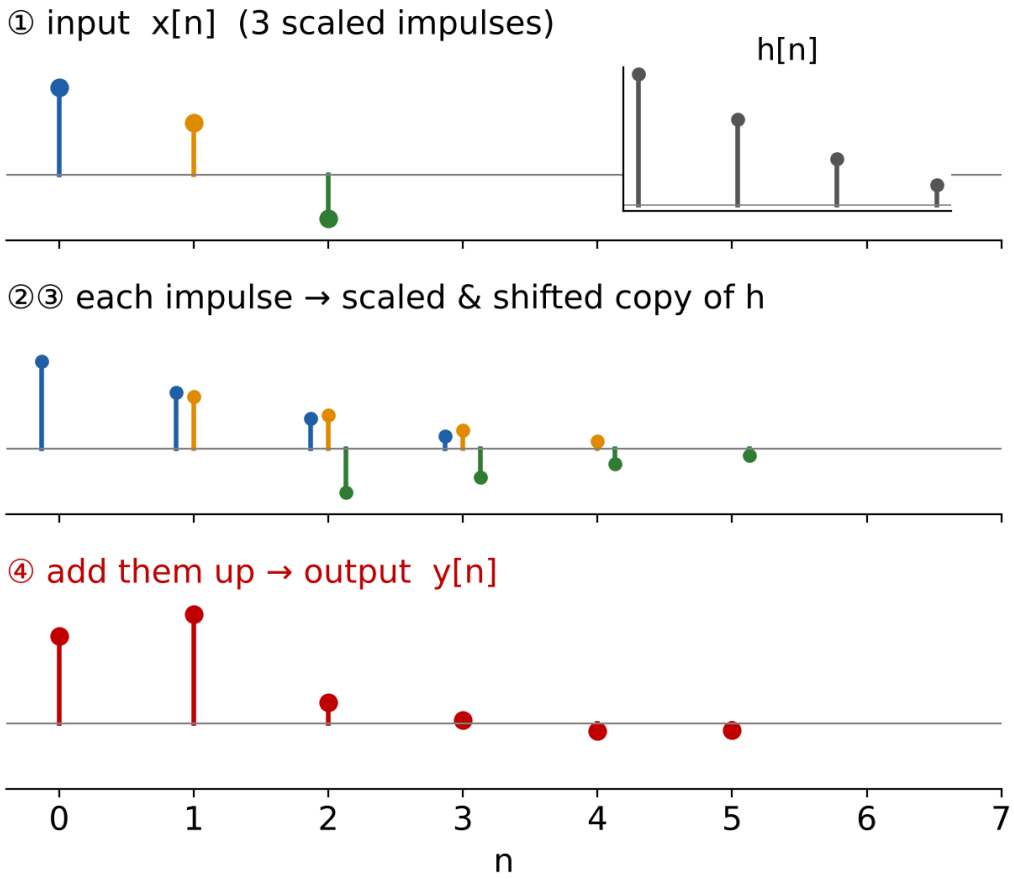
① Split $x[n] = \sum_k x[k]\,\delta[n-k]$

② Time-inv. $\delta[n-k] \;\rightarrow\; h[n-k]$

③ Linearity $x[k]\,\delta[n-k] \;\rightarrow\; x[k]\,h[n-k]$

④ Sum $y[n] = \sum_k x[k]\,h[n-k] = (x * h)[n]$

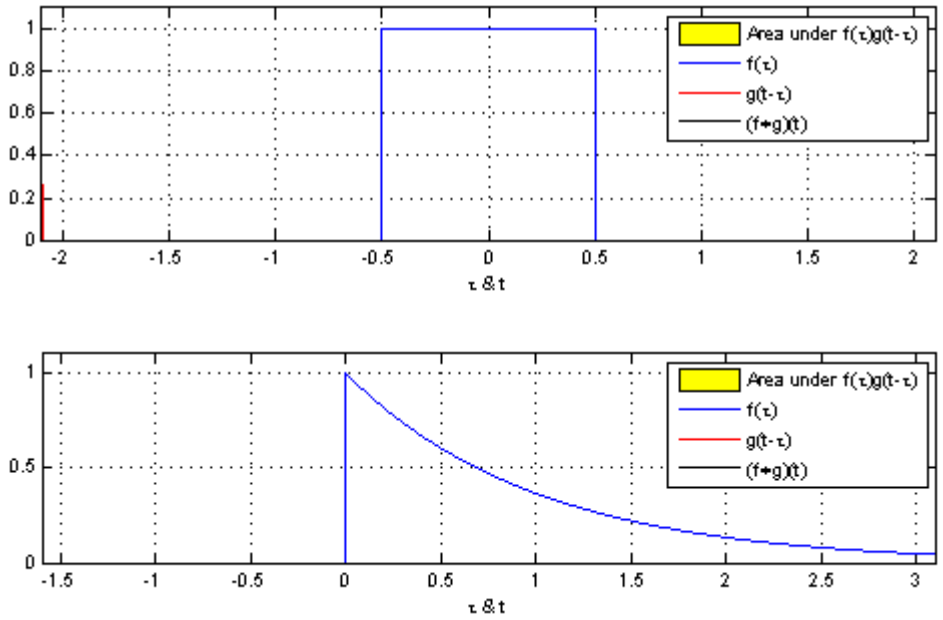
FFT $Y(\omega) = H(\omega)\,X(\omega)$ |H|: gain, ∠H: delay

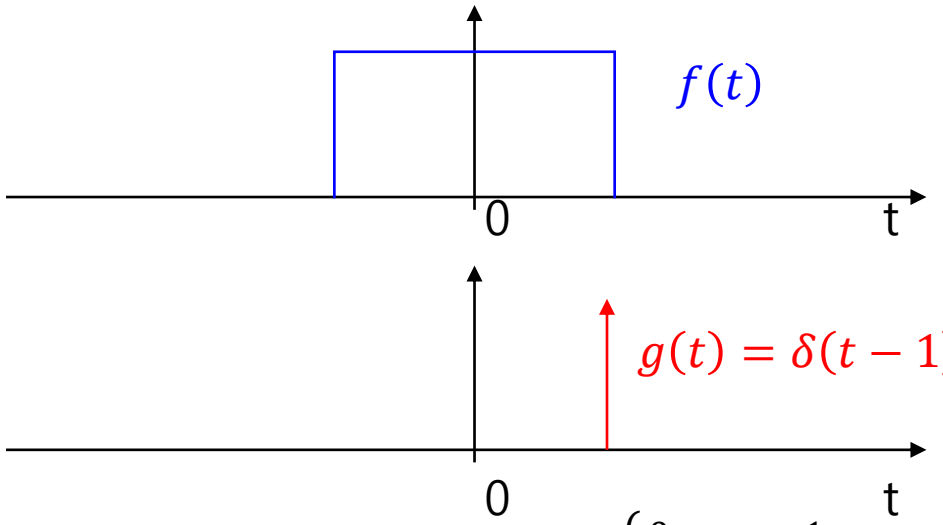


Multi-channel Source Separation

- Microphone Array
  - Convolution: $(f * g)(t) := \int_{-\infty}^{\infty} f(\tau) g(t-\tau)\, d\tau.$

Example of Convolution





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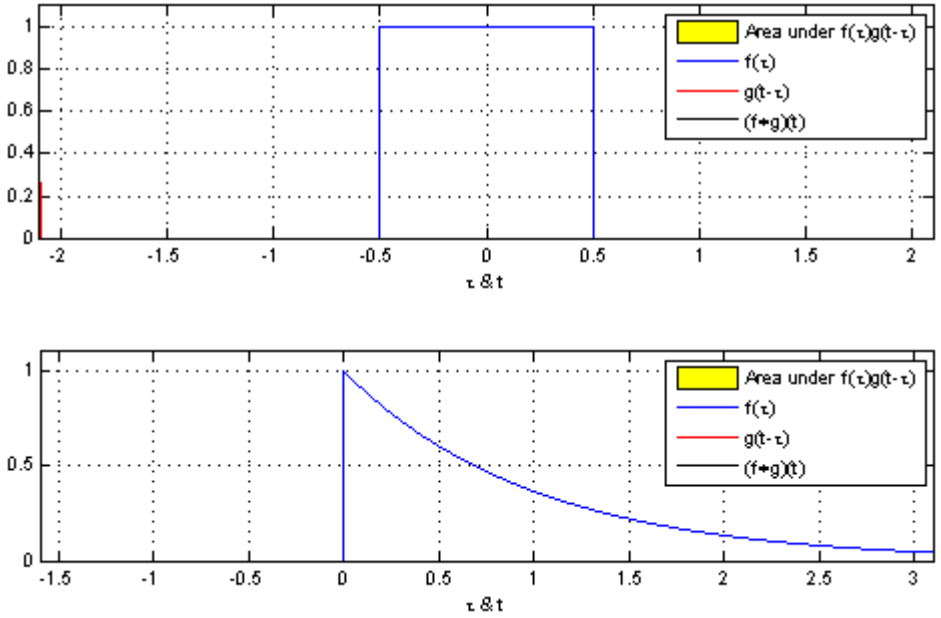
$$\int_{-\infty}^{\infty} \delta(t-1)dt = 1$$

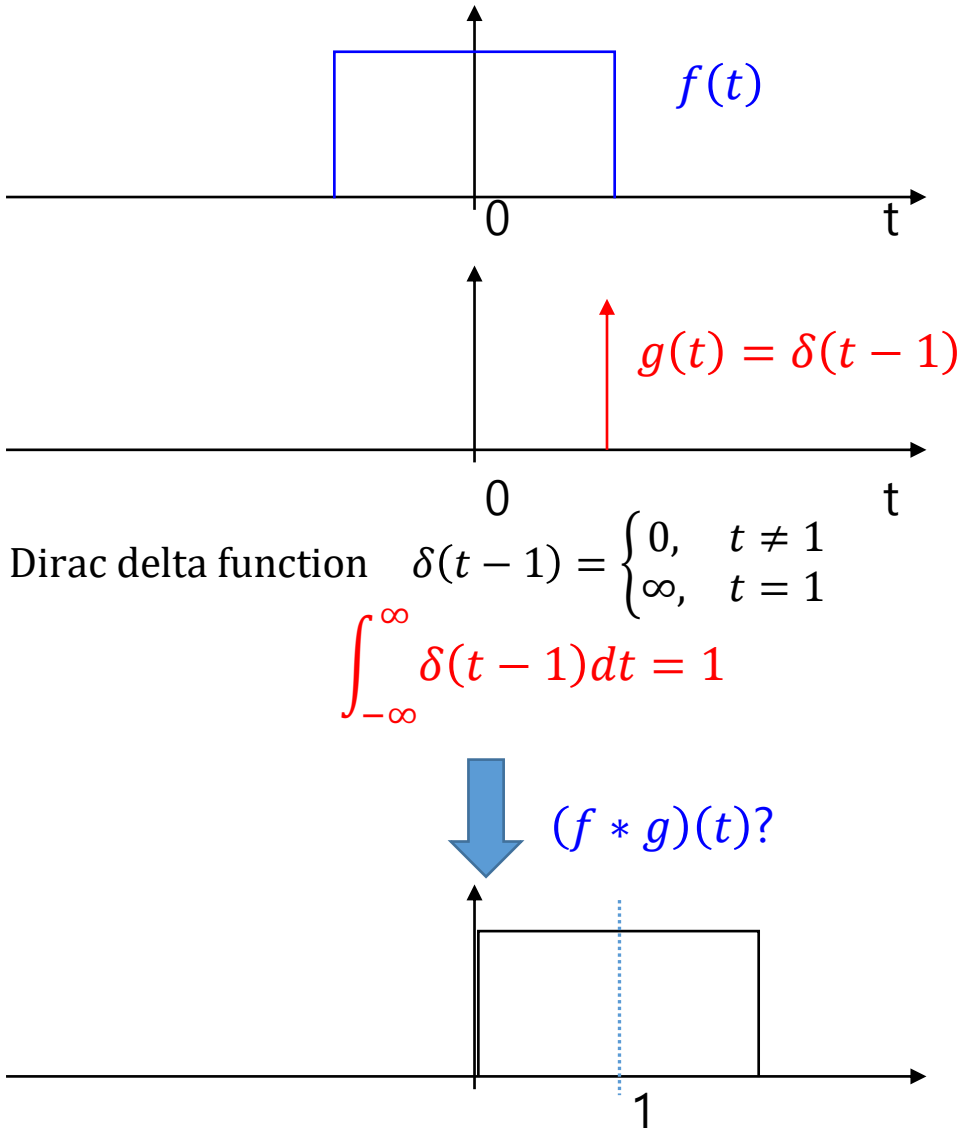
$(f * g)(t)$?

Multi-channel Source Separation

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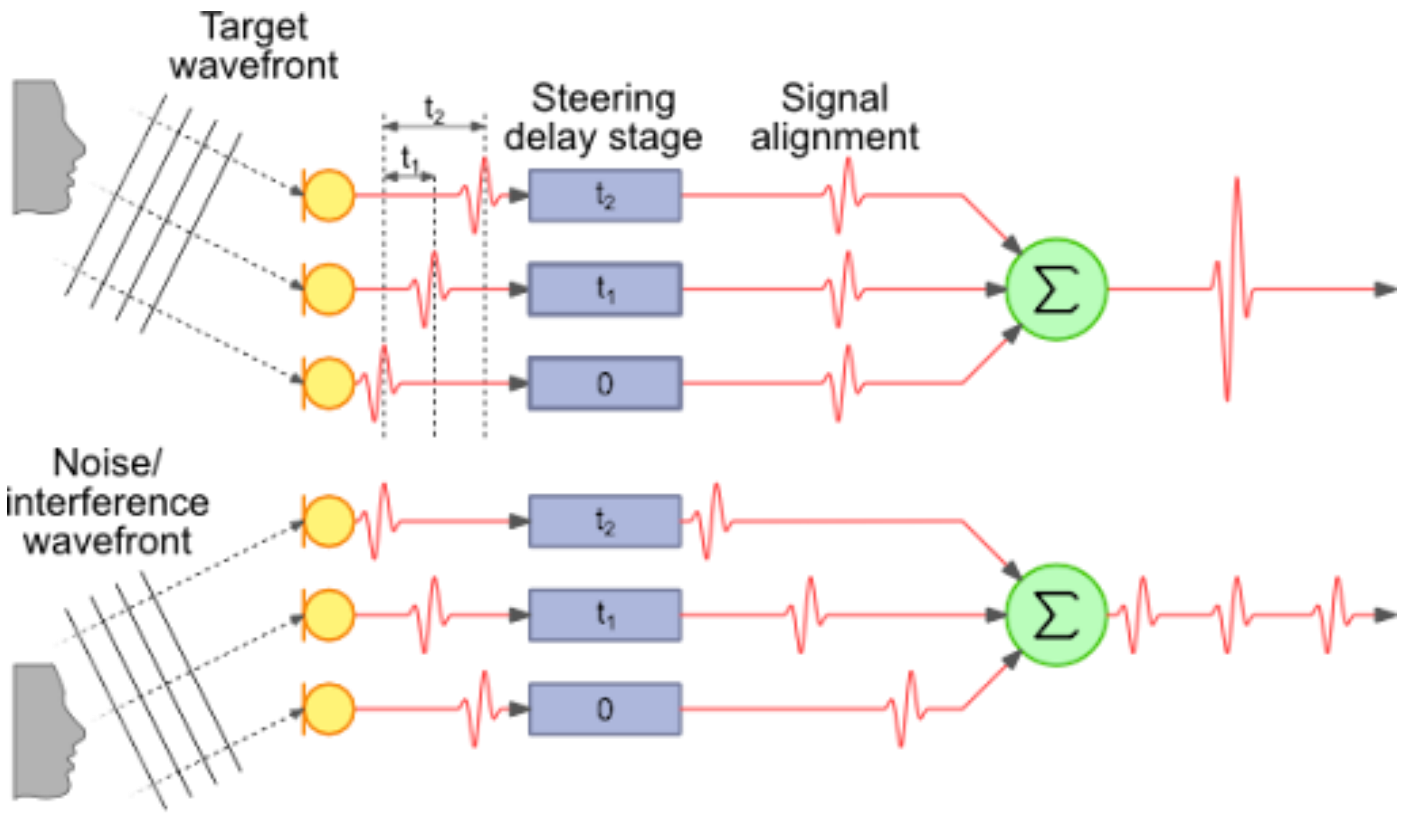
Example of Convolution





Multi-channel Source Separation

- Delay-and-Sum Beamformer
  - Consider the propagation delays!



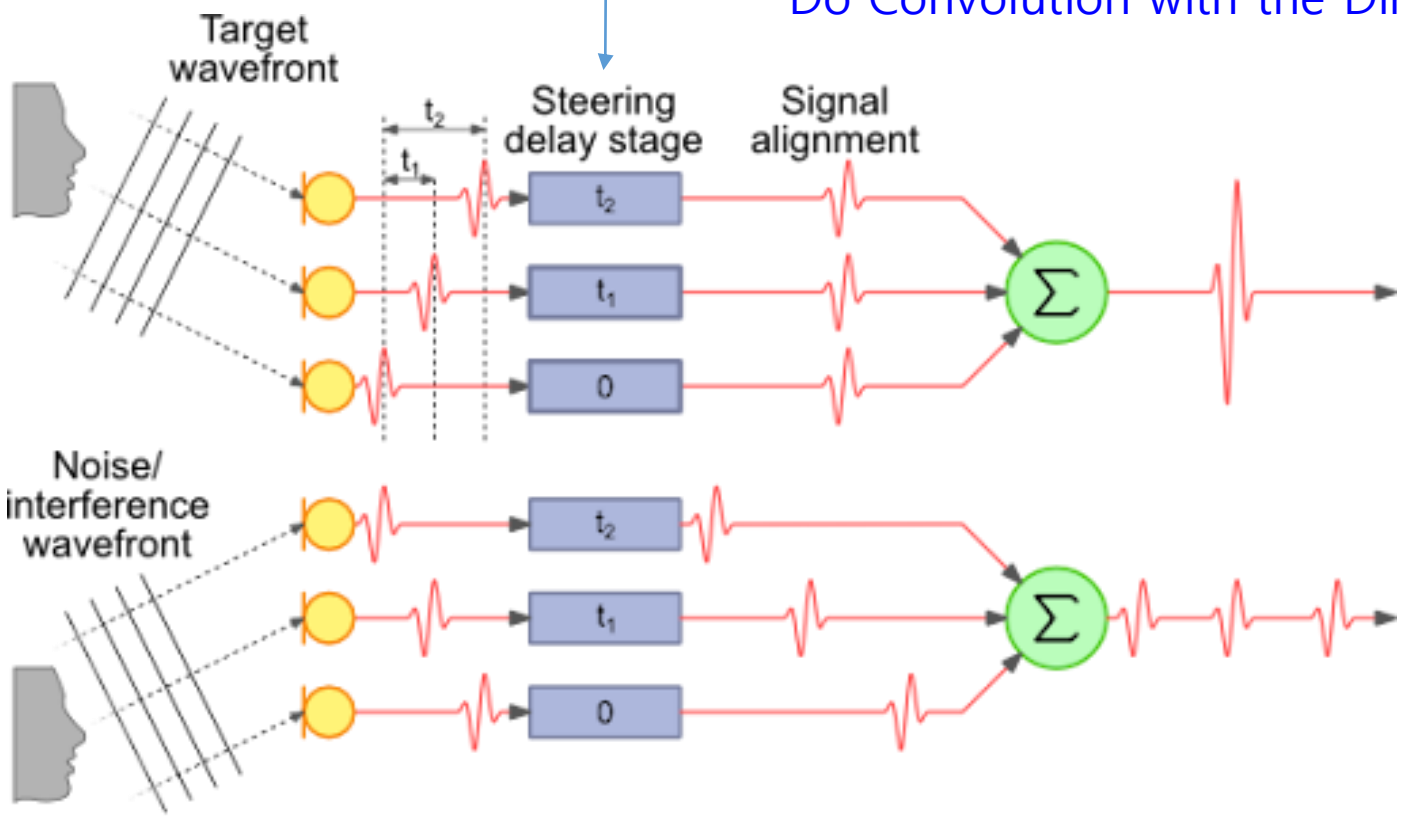
https://medium.com/@itberrios6/introduction-to-beamforming-part-1-92767408f483

Multi-channel Source Separation

- Delay-and-Sum Beamformer
  - Consider the propagation delays!

$$g(t) = \delta(t + t_1)$$

Do Convolution with the Dirac delta function!



https://medium.com/@itberrios6/introduction-to-beamforming-part-1-92767408f483

Multi-channel Source Separation

- Microphone Array
  - Is it possible to extract only the audio signal coming from a specific direction? → **Beamforming**
  - What is the beamforming algorithm?
    - Focusing a specific direction, $\boldsymbol{\theta}$, by using beamforming weights (spatial filter)

$$y(\theta,t)=\sum_{m=1}^{M} g_m(\theta,t) * x_m(t) \quad \Longleftrightarrow \quad y(\theta,\omega)=\boldsymbol{w}(\theta,\omega)^H\,\boldsymbol{x}(\omega)$$

Convolution

FFT & iFFT

Beamforming output: $y(\theta,\omega)$ — Weight (Spatial filter): $\boldsymbol{w}(\theta,\omega)^H$ — Measured pressure signals: $\boldsymbol{x}(\omega)$

$$\boldsymbol{\delta(t+t_1)} \quad \Longleftrightarrow \quad e^{+j\omega t_1}=w_1^*(\omega)$$

FFT & iFFT

$$\Rightarrow\ \boldsymbol{w}(\theta,\omega)=\frac{1}{M}[e^{-j\omega t_1},\dots,e^{-j\omega t_M}]^T,\quad y(\theta,\omega)=\boldsymbol{w}^H\boldsymbol{x}(\omega)=\frac{1}{M}\sum_{m=1}^{M} e^{+j\omega t_m}\,X_m(\omega)$$

Multi-channel Source Separation

- Steering vector: a time delay becomes a phase shift

Delay

$$\tau_m(\theta) = \frac{(m-1)\, d \sin\theta}{c}$$

extra travel time to mic m (plane wave, mic spacing d)

Steering vector

$$\boldsymbol{h}(\theta, \omega) = [e^{-j\omega\tau_1(\theta)}, \ \ldots, \ e^{-j\omega\tau_M(\theta)}]^T$$

what the array receives from direction θ: x(ω) = h(θ, ω)·S(ω)

DAS

$$\boldsymbol{w}_{\mathrm{DAS}} = \frac{\boldsymbol{h}}{M}, \qquad y = \boldsymbol{w}^H \boldsymbol{x} = \frac{1}{M} \sum_{m=1}^{M} e^{+j\omega\tau_m} X_m(\omega)$$

why h / M ? → next slide

Steered to the source (θ = θ₀): $\mathbf{w}^H\mathbf{h} = 1$ → y(ω) = S(ω)

Multi-channel Source Separation

- Why can the steering vector itself be the DAS filter?

① The array receives delays (= steering vector)

$$X_m(\omega) = e^{-j\omega\tau_m} S(\omega) = h_m\, S(\omega)$$

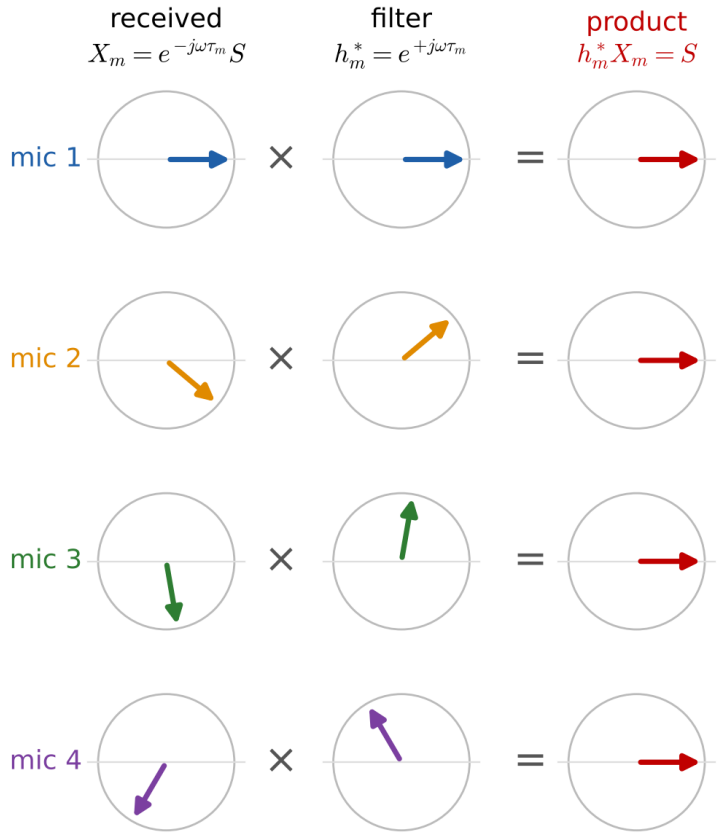
② DAS undoes each delay

$$g_m(t) = \delta(t + \tau_m) \overset{\text{FFT}}{\Leftrightarrow} G_m(\omega) = e^{+j\omega\tau_m} = h_m^*$$

undoing a delay = multiplying by the conjugate phase

③ Sum over the mics = $w^H x$

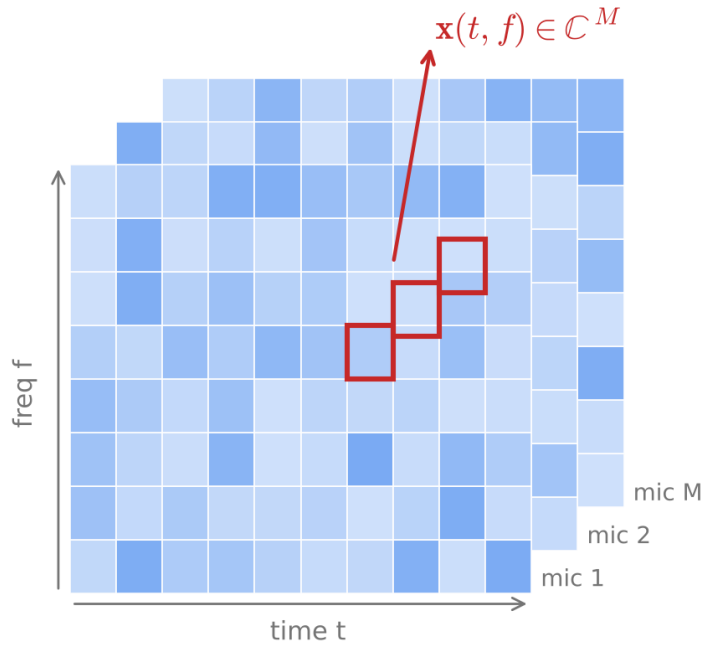
$$y = \frac{1}{M}\sum_{m=1}^{M} h_m^*\, X_m = \left(\frac{\boldsymbol{h}}{M}\right)^H \boldsymbol{x} \quad \Rightarrow \quad \boldsymbol{w}_{\text{DAS}} = \frac{\boldsymbol{h}}{M}$$



The filter that undoes the delays is h*, and w^H already takes the conjugate → w = h

Multi-channel Source Separation

- Why the STFT? Convolution becomes multiplication per bin



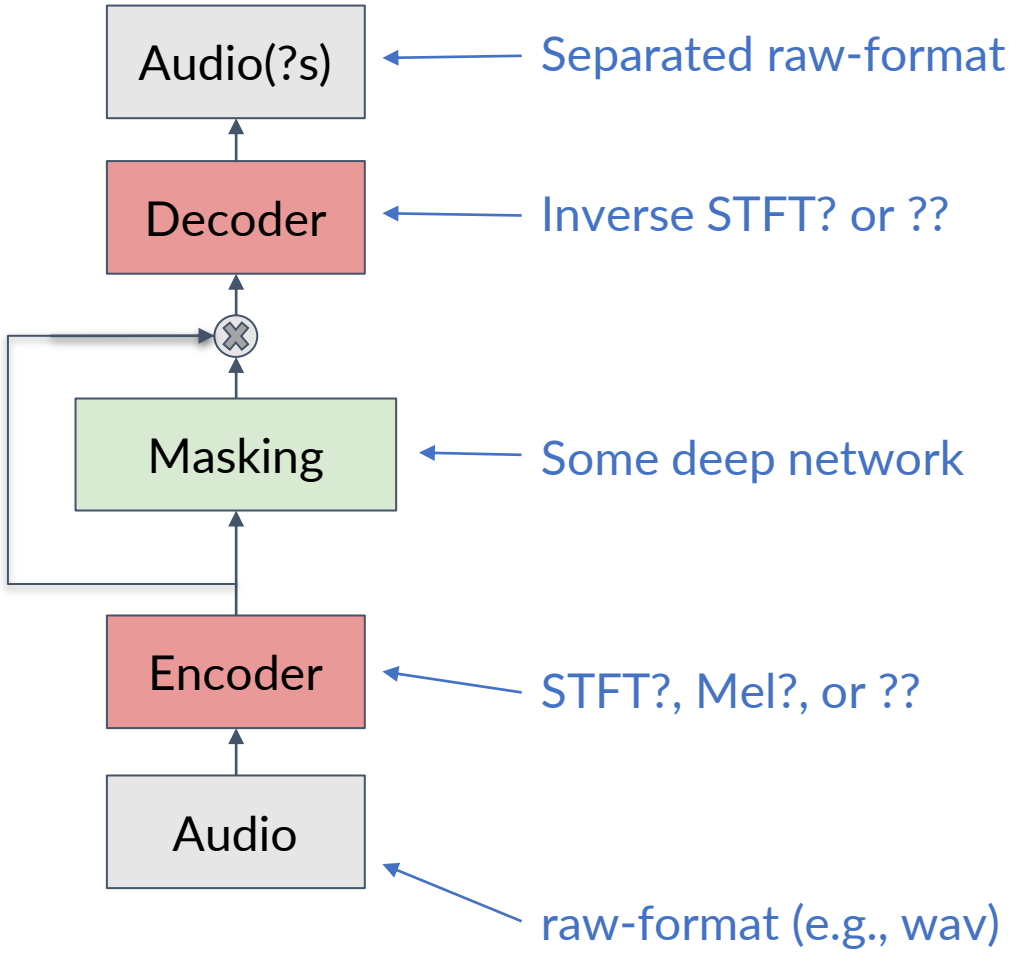
Room (time) $x_m(t) = (a_m * s)(t)$

STFT bin $\boldsymbol{x}(t, f) \approx \boldsymbol{a}(f)\, S(t, f)$

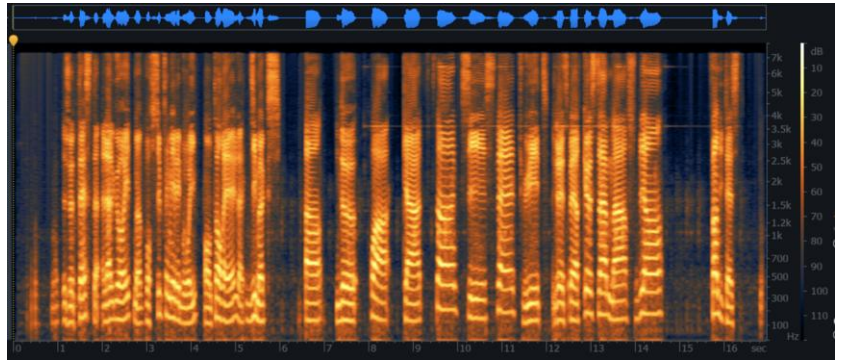
Filter per bin $y(t, f) = \boldsymbol{w}(f)^H \boldsymbol{x}(t, f)$

ESD = STFT → per-bin spatial filter → iSTFT

Encoder-Separation-Decoder (ESD)

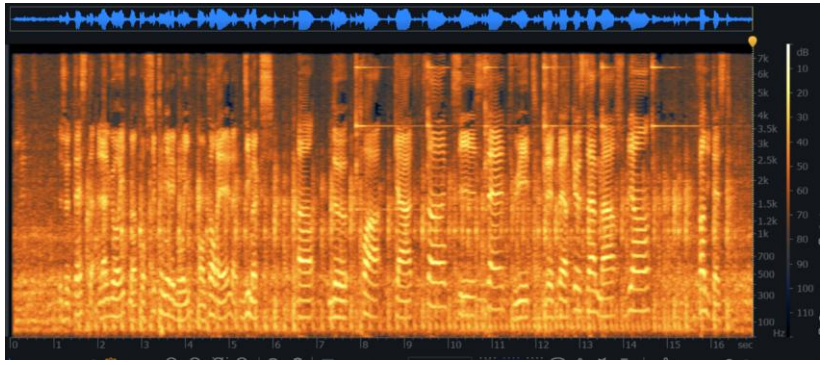


Denoised Alex

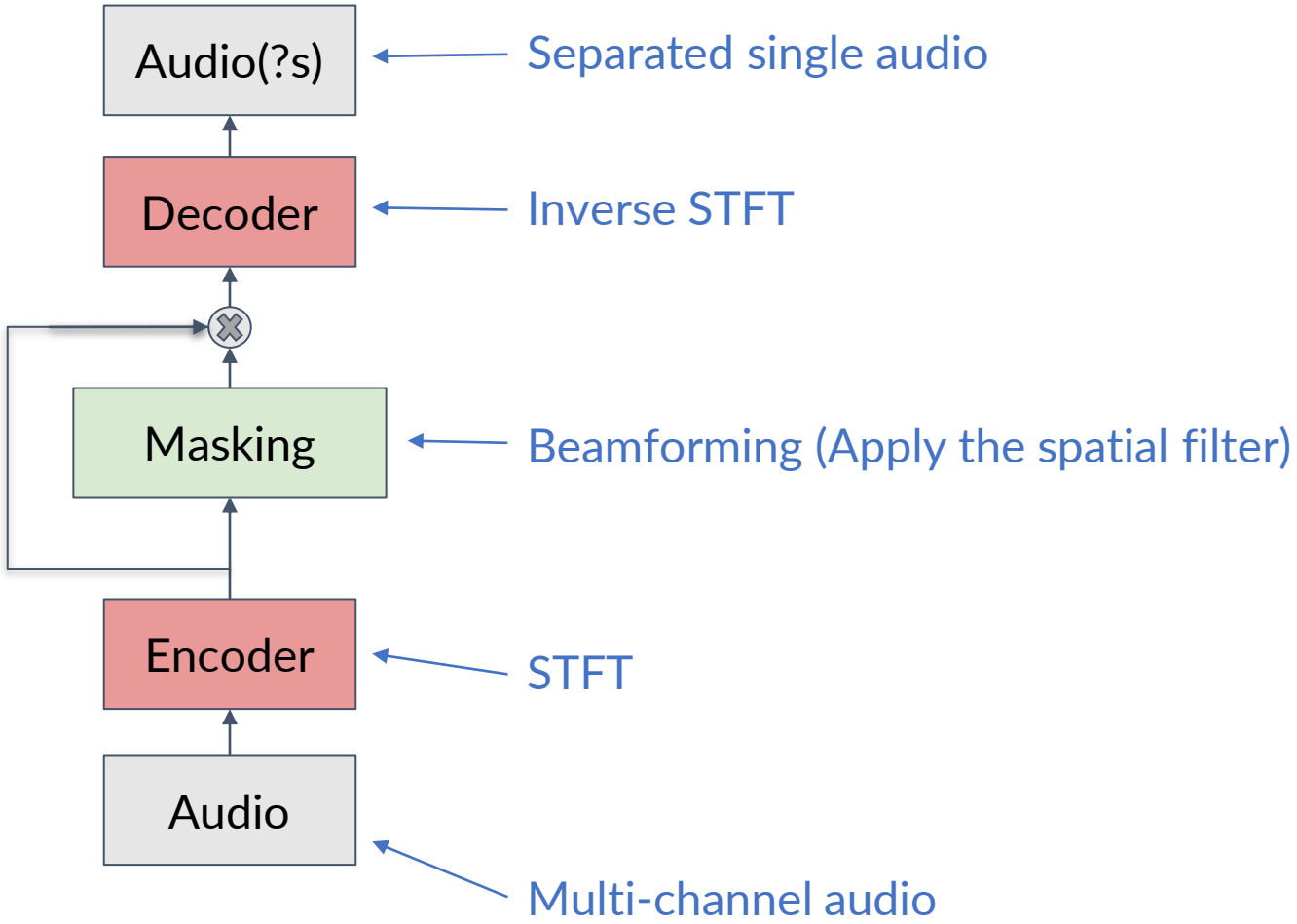


Denoising

Alex & noise

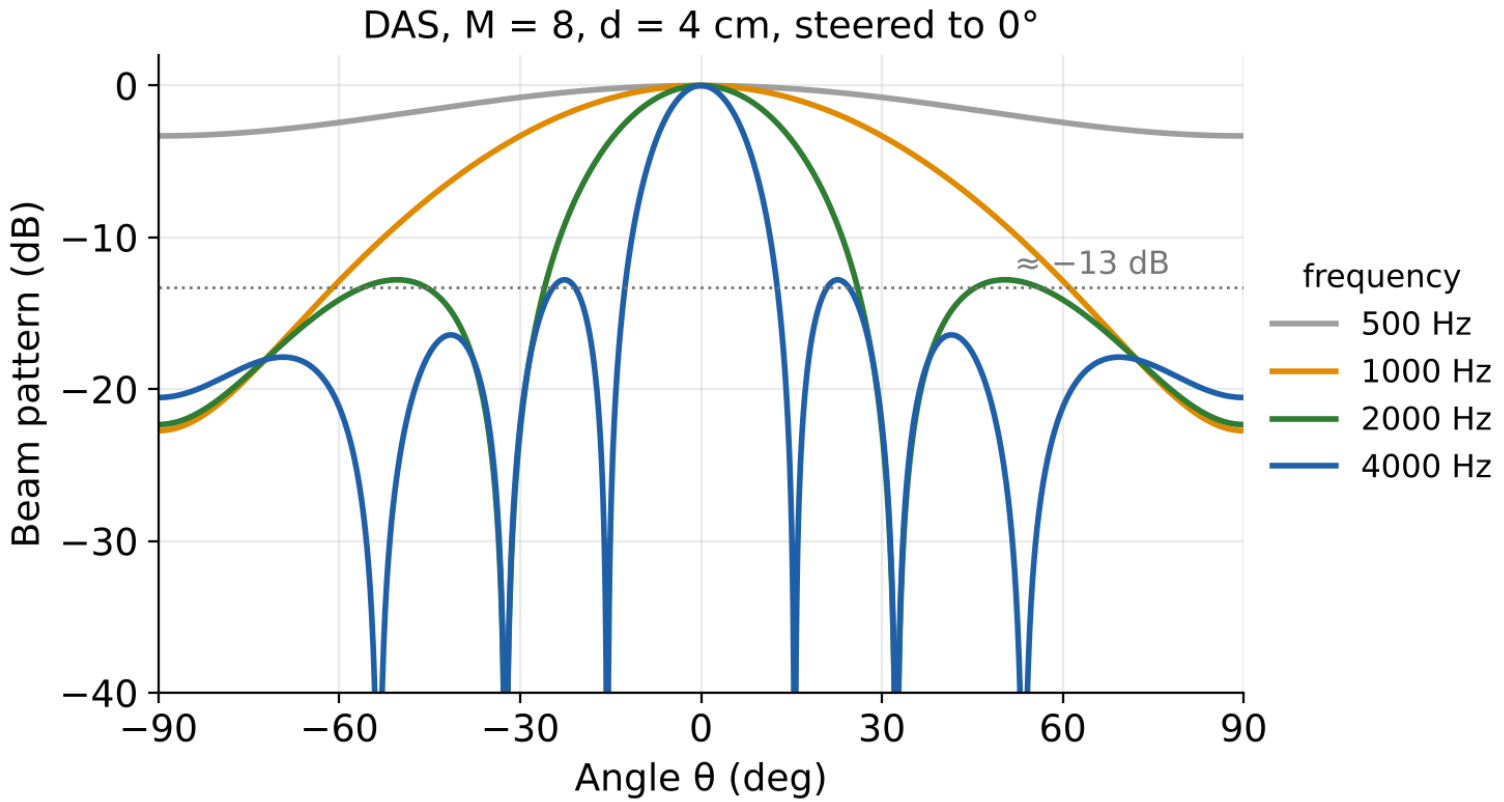


Multi-Channel Encoder-Separation-Decoder (ESD)



Multi-channel Source Separation

- DAS beam pattern: sharper at high frequency



$$B(\theta) = |\boldsymbol{w}_{\mathrm{DAS}}^{H}\,\boldsymbol{h}(\theta)|^{2}$$

Main-lobe width ∝ λ / (M·d)

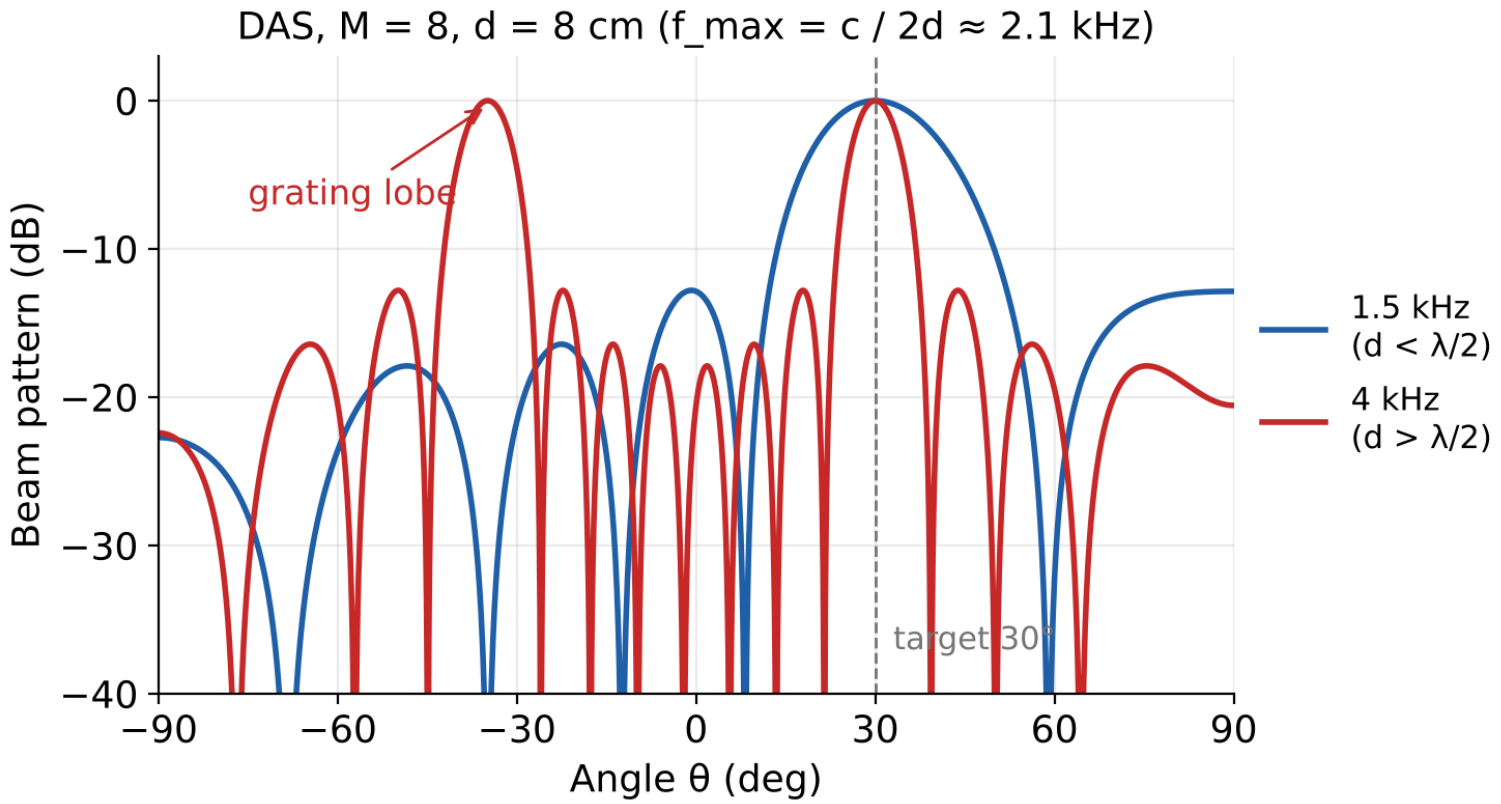
→ narrow at high f, wide at low f

Side lobes ≈ −13 dB

→ fixed for uniform weights

Multi-channel Source Separation

- Spatial aliasing: microphones too far apart



$$d \leq \frac{\lambda}{2} = \frac{c}{2f}$$

d > λ/2 → grating lobe

4 kHz: a 2nd full peak at −35°

d = 8 cm → f_max ≈ 2.1 kHz

d = 4 cm → f_max ≈ 4.3 kHz